

Lipolyse-C

The Most Potent Hydrophilic Emulsifier

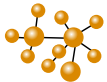


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OIL INCLUSION IN BROILER IS IMPORTANT, WHY?

Lipids constitute the main energy reserve of birds and it has the highest caloric value among all nutrients. The Carbon atoms of the fatty acids are chemically more than C atoms found in Sugar. Therefore, the oxidation of triglycerides releases more than twice as much energy as Carbohydrates.

The deposition of 1 g of energy from carbohydrate (or) protein by a poultry bird requires higher quantities of these nutrients in comparison to the deposition of 1g of energy from fat.



COMPOSITION

- GPGR (Glyceryl Poly Ethylene Glycol Ricinoleate)
- Lysophospholipid
- Phospholipids
- Soluble Caseinates
- Lipase
- Synthetic Bile Salt
- Natural Carriers



HOW DOES AN EMULSIFIER WORKS?

Emulsifier molecules work by having a hydrophilic end (water-loving) and hydrophobic end (water-hating). The hydrophilic end of the emulsifier molecule is attracted to the water and the hydrophobic end is attracted to the fat/oil. By vigorously mixing the emulsifier with the water and fat/oil, a stable emulsion can be made.

Fat emulsification in the digestive tract occurs with the help of bile salts and later digested by Lipase. But, the most critical step is digestion of Fatty acid containing Carbon atoms >14. As most of the oil included in the broiler diet has fatty acid with >14 Carbon atoms, further addition of an emulsifier in feed becomes mandatory .

EMULSIFIERS ARE REQUIRED IN :



YOUNG BIRDS: Young birds have limited ability to digest fats which is due to limited bile salt secretion and absorption, which increases only with age. So, In broilers the addition of emulsifiers plays major role in production cost.

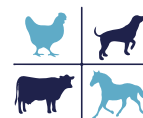
DISEASE CONDITION: Pathogenic organism affects intestinal epithelium. Epithelial cells are ruptured by which absorption of essential nutrients are decreased.



CEREALS HAVING HIGH LEVEL OF NSPs: High level of NSPs increases the viscosity of digesta in chicken intestine and interferes in absorption of nutrients. Increased viscosity also leads to impaired peristaltic progress of chyme through intestine which leads to bacterial overgrowth, causing loss of conjugation of bile salts and reduction of fat utilisation.

All the above factors causes poor growth in broiler chickens with great loss to farmers . So, it becomes mandatory to add emulsifiers in the diet of broiler ration.

Glyceryl Poly Ethylene Glycol Ricinoleate (GPGR)



- ✦ More hydrophilic and dissolves the free FA which are hardly soluble in bile salt, enhances dietary fat availability, improves body weight and relative organ weight.
- ✦ GPGR maintains serum and meat lipid profile.
- ✦ Adding GPGR in feed before pelleting increases humidity and decreases pellet press energy consumption and improves the pellet quality by modulating the moisture content during pelleting process.

LYSOPHOSPHOLIPID or LYSOLECITHIN:

- ✦ Lysophospholipid improves body weight gain, FCR, improves carcass quality and dressing percentage. Choline is mainly meant to reduce fatty liver condition in birds and promotes jejunal function in broilers.
- ✦ It acts as an agonist of PPAR α (Peroxisome Proliferator Activated Receptors Alpha), proteic nuclear receptors acting on transcription process. The PPAR α regulate the expression of specific genes implicated in metabolic reactions, such as β -oxidation of fatty acids.
- ✦ Lysophospholipid are modulators of epigenetic regulation of gene expression, which improves fatty acid digestion in the birds.
- ✦ Acts as a methyl group donor via betaine, helps in synthesis of phospholipids and maintains cell wall integrity.

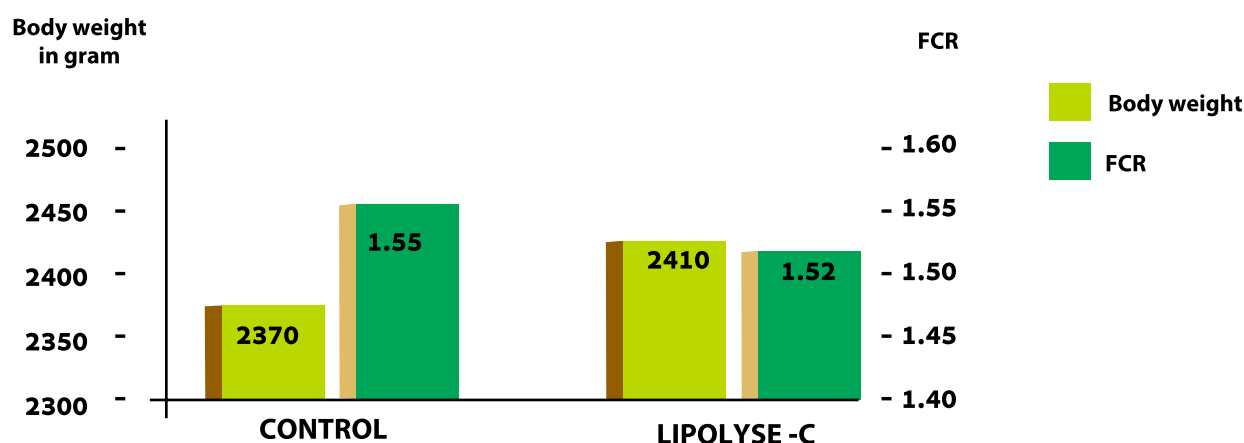
SOLUBLE CASEINATES:

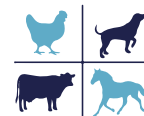
- ✦ A potent emulsifying agent due to its amphiphilic property. It combines with both water and oil phase and stabilizes the emulsion. Increases absorption of proteins & fatty acids, improves body weight gain and increases lipase activity.

SYNTHETIC BILE SALT

- ✦ Added Synthetic Bile Salt in LIPOLYSE-C activates lipase enzyme, increases ME, improves digestion of fat molecules, decreases plasma cholesterol.

Effect of LIPOLYSE -C supplementation on FCR & body weight of broilers at 42 days of age





LIPASE

Lipases are defined as triacylglycerol acyl hydrolases that are involved in the hydrolysis of fats and oils to yield glycerol and free fatty acids.. Young birds with immature digestive capability for complete absorption of lipids could particularly benefit from dietary lipases (Pleiss et al., 1998; Fickers et al., 2011).

HYDROPHILIC LIPOPHILIC BALANCE:

The extent to which the part of an emulsifier are hydrophilic & lipophilic in nature determines HLB value. HLB is the key indicator for selecting an emulsifier, ranging from 0-20 . As, in the gut it is Oil-Water emulsion high HLB is required.



KEY BENEFITS:

- Increases the digestibility of fat & oil in the feed.
- Increases the digestion & absorption of proteins in the diet.
- Facilitate low cost feed formulation.
- Minimizes abdominal fat .
- Enhances the secretion and activities of lipase.
- Maximizes enzymatic digestion of feed.
- Suitable for pelleting.



DOSAGE

Broiler/Layer : 500 g/ton of feed.

or

Advised by the Veterinarian



PRESENTATION

25 KG Bag



SHELF LIFE

24 Months from Date of Manufacture.

STORAGE

Keep it in a cool place.



(An ISO 9001 : 2015 Certified Company)



GMP & HACCP Certified

Manufactured and Marketed By :

ZENMAK NUTRIGENCIES AND HEALTH PVT. LTD.

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